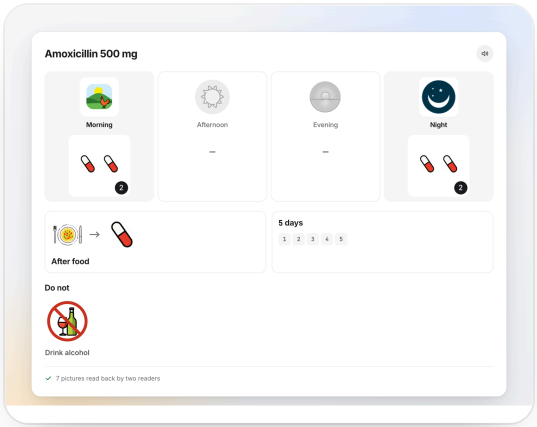
Visual discharge instructions, verified before the patient sees them.

A discharge sheet, typed or photographed, becomes picture cards a patient can follow. An image model draws every pictogram. Two other models have to read each one back correctly before it reaches a card.



Live concord-health.vercel.app, no sign-in

Code github.com/kinnuworks/concord

Video under 3 minutes, linked on the submission page

DOIs 10.5281/zenodo.23346534 (report), 10.5281/zenodo.23346012 (code)

Where to look, by judging criterion

01
Technical execution

A five-step pipeline with our own fine-tune. Two models from different companies read the sheet, and any field they disagree on is flagged. Code lays out each card. FLUX.1 [dev] with a LoRA adapter we trained draws one fact per picture. Two vision models then describe the picture with no hint, a separate judge compares, and a misread picture is redrawn up to three times.

See `src/lib/gate.ts` `src/lib/readback.ts` `scripts/train.ts`

02
Innovation

A generated picture has to pass a reading test. Readers that were told nothing must say in their own words what it shows, with no list of answers to pick from. And the image model is never asked to count: it draws one pill, the check confirms it is exactly one, and code repeats it.

03
Use of image AI

Every pictogram on a card is generated, and the design follows from how an image model behaves. Our adapter, trained on 70 open pictograms, took the share of pictures drawn in one flat pictogram style from 21% to 78%, so a set of cards shares one look.

04
Impact

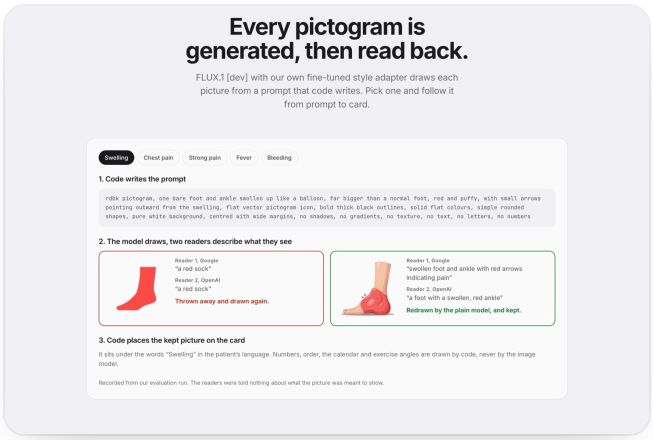
Shown the label "Take two tablets by mouth twice daily", 71% of low-literacy patients could say it back and 35% could count out the pills (Davis et al., 2006). Picture instructions cut dosing errors from 47.8% to 5.4% in a randomized trial (Yin et al., 2008). Concord serves the people text leaves behind: 61.2 million Americans aged 65 or older and 28.9 million who speak English less than very well (Census Bureau, 2024), in 70 languages.

05
Design and demo

The site opens on finished cards. A photographed sheet turns into cards in seconds, the language switches in one click, and the cards print on two clean pages. Lighthouse scores the live site 100 in all four categories.

Try it in two minutes

1. Open **concord-health.vercel.app**. Finished cards are already on screen.
2. Click **A photo of a printed sheet**, then **Make the cards**. The sheet is read live by two models.
3. Pick any of the **70 languages**, such as Hindi or Arabic. The medicine's name appears in that script under the printed one. Press the speaker, then **Print the cards**.
4. Scroll to **Where the image model works** and click through the five examples. Each shows the prompt, a picture the readers misread, and the one that was kept.



A recorded run. Asked for a swollen foot, the model drew what both readers called "a red sock". It was thrown away and drawn again.

What we measured

83 of 168 An image model on its own whole-dose pictures showed the wrong number of pills when the model was asked to draw the dose in one picture.

58 of 60 With Concord single-pill pictures passed the read-back on the first try. Code repeats the checked pill, so the count on a card is exact.

21% to 78% With our adapter pictures drawn in one flat pictogram style, 270 each way. This measures style; every picture on a card has still passed the read-back.

155,637 With Concord medicine names read back by a second model that never saw the English: 1,830 prescribable medicines in each of 69 languages, plus 29,367 names from Wikidata.

115,200 With Concord different medicine cards can be built from 16 checked pictures, and the count on every one is exact because code repeats one checked pill.

100 of 100 With Concord written test instructions read correctly by the two sheet readers.

540 pictures were drawn for a fixed set of 100 instructions and every one was read by both readers. The scripts are in `scripts/` and the raw readings in `training/eval/`.

Built during the hackathon

- A LoRA style adapter for FLUX.1 [dev], trained on 70 OpenMoji pictograms.
- The read-back check: two readers, a judge, redraws, and a code-drawn pictogram when nothing passes.
- Cards in 70 languages, with 22,199 medicine names taken from Wikidata and every name read back.
- 29 automated tests on every commit, and 0 accessibility violations on the live site.